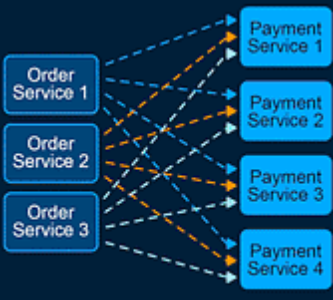


Top 9 System Integrations

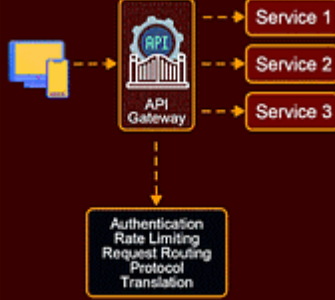
ByteByteGo

luglio
2024

Peer-to-Peer



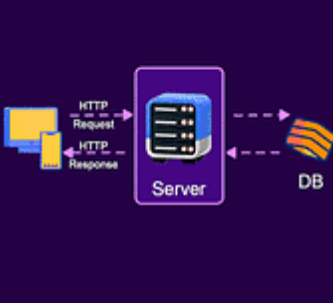
API Gateway



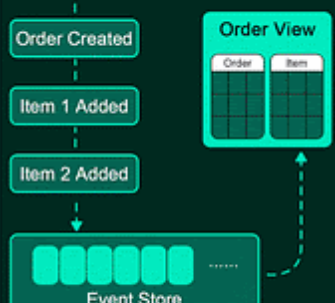
Pub-Sub



Request-Response



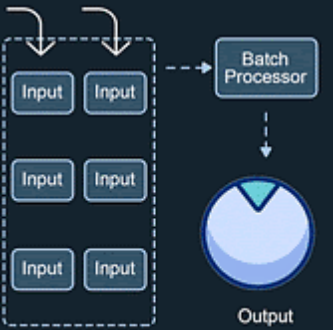
Event Sourcing



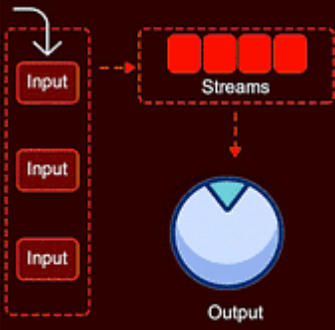
ETL



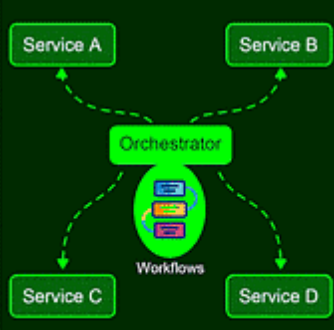
Batching



Streaming Processing



Orchestration



- Peer-to-Peer (usually synchronous)

The Peer-to-Peer pattern involves direct communication between two components without the need for a central coordinator.

- API Gateway (usually real-time) / SOA (Service Oriented Architecture)
An API Gateway acts as a single entry point for all client requests to the backend services of an application.

- Pub-Sub (asynchronous)
The Pub-Sub pattern decouples the producers of messages (publishers) from the consumers of messages (subscribers) through a message broker.

- Request-Response (synchronous)
This is one of the most fundamental integration patterns, where a client sends a request to a server and waits for a response.

- Event Sourcing (real-time)
Event Sourcing involves storing the state changes of an application as a sequence of events.

- ETL (batch – usually during the night or the week-end)
ETL is a data integration pattern used to gather data from multiple sources, transform it into a structured format, and load it into a destination database.

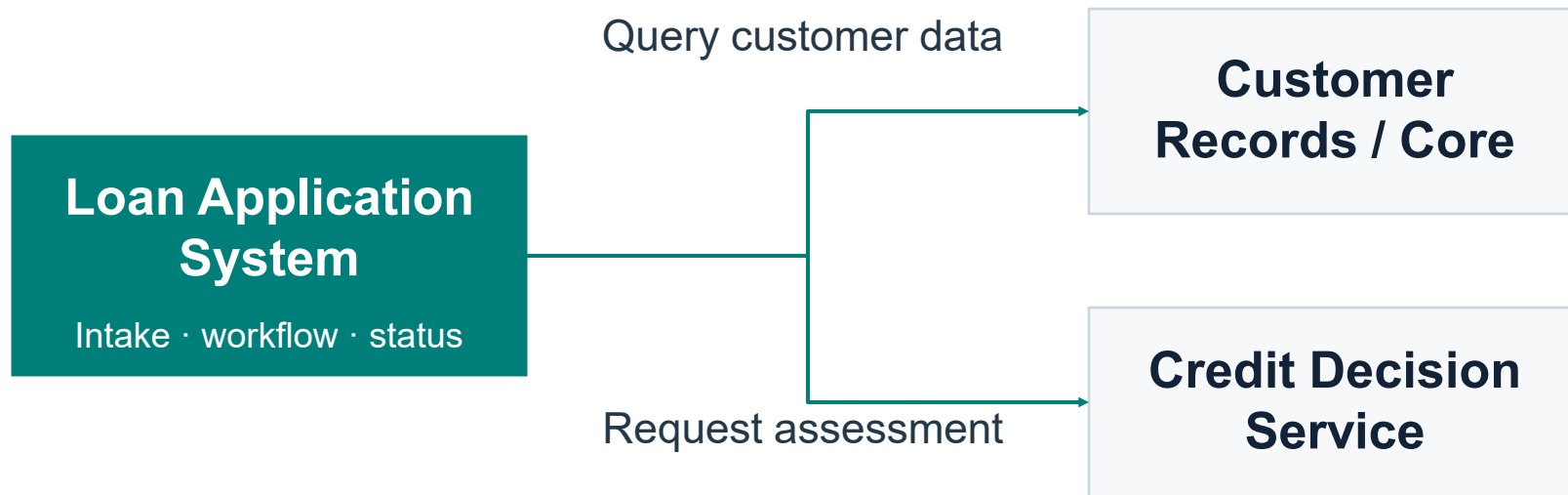
- Batching (many bank batches that run often during the night)
Batching involves accumulating data over a period or until a certain threshold is met before processing it as a single group.

- Streaming Processing (eg, images from the bank's caveau or the bank access doors; other example: financial info about stock exchanges)
Streaming Processing allows for the continuous ingestion, processing, and analysis of data streams in real-time.

- Orchestration (BPM = Business Process Management / Workflow)
Orchestration involves a central coordinator (an orchestrator) managing the interactions between distributed components or services to achieve a workflow or business process.

Peer-to-Peer

Direct point-to-point links connect selected systems without a shared integration intermediary.



ILLUSTRATIVE EXAMPLE

Loan intake queries customer records and requests assessment from the Credit Decision Service.

GOOD FIT

Few, stable interfaces with clear owners.

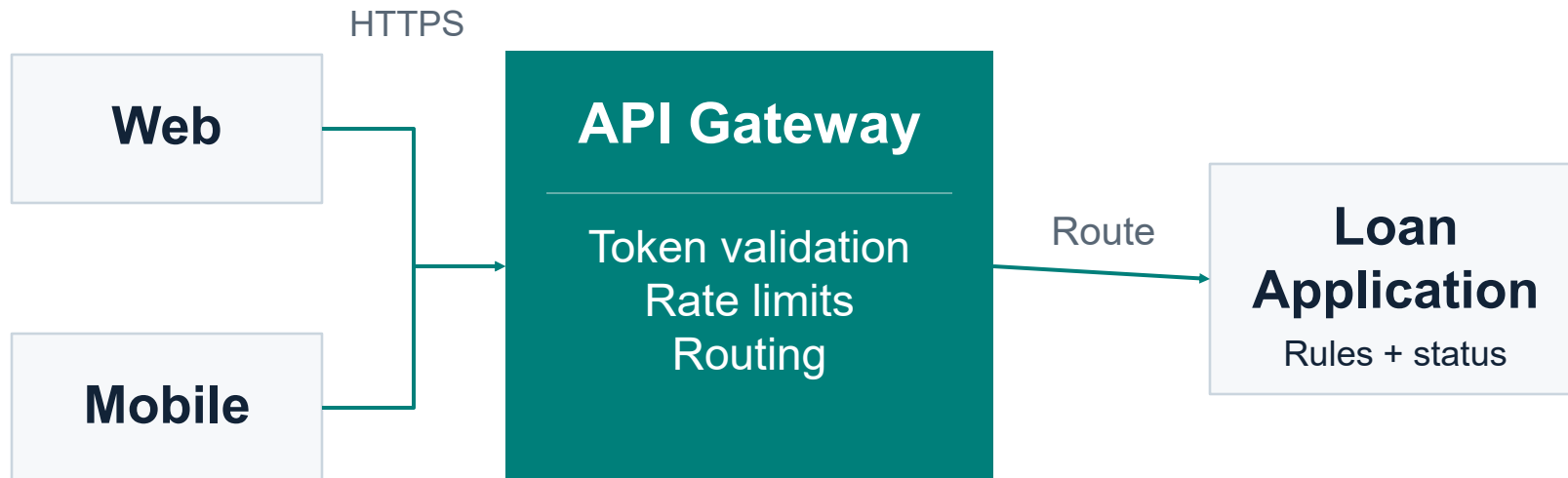
Direct requests are shown. Responses and unrelated connections are omitted.

DESIGN WATCHPOINT

Each connection needs an owned contract, security, error handling and monitoring. More links can increase maintenance effort.

API Gateway

A gateway presents an API entry point, routes requests to backends and applies shared edge policies.



The gateway handles entry policies. Domain services own business rules.

ILLUSTRATIVE EXAMPLE

Web and mobile customers access loan APIs through a common entry point.

GOOD FIT

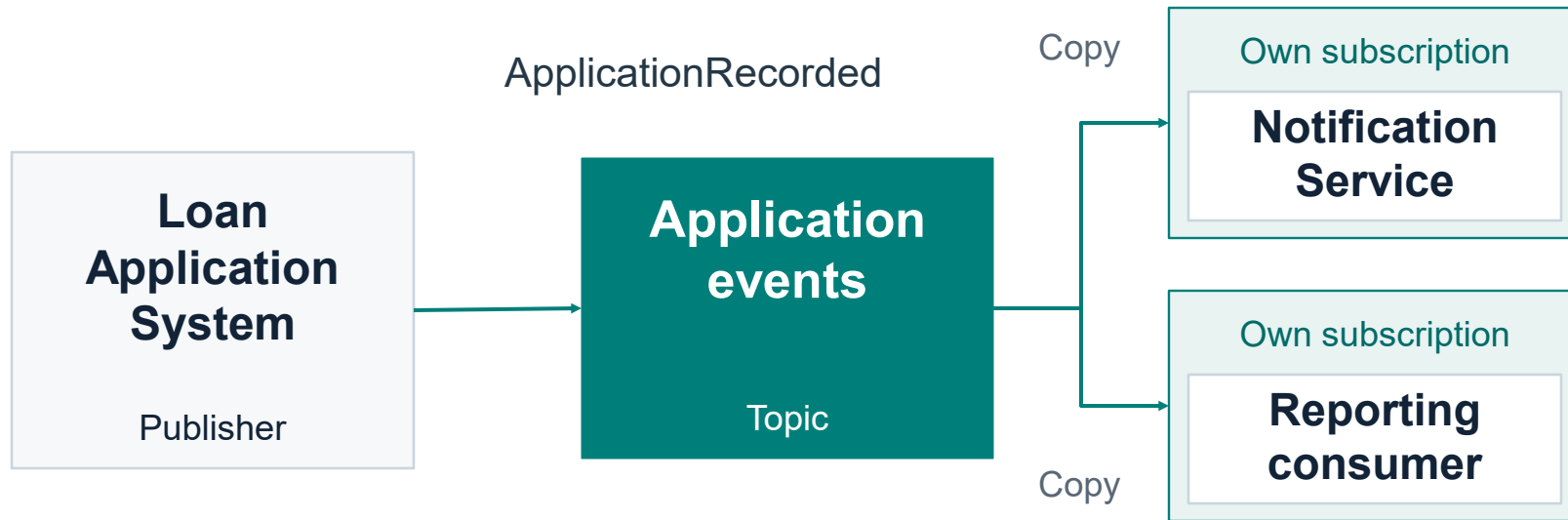
Consistent entry policies across multiple clients and APIs.

DESIGN WATCHPOINT

Plan gateway capacity and availability. Keep business authorization in domain services and credit decisions in the Credit Decision Service.

Publish-Subscribe

A publisher emits a message to a topic; matching messages are made available independently to subscribers through their own subscriptions.



Each subscription keeps its own delivery state; replicas can share work within it.

ILLUSTRATIVE EXAMPLE

ApplicationRecorded feeds notification and reporting consumers after the application is saved.

GOOD FIT

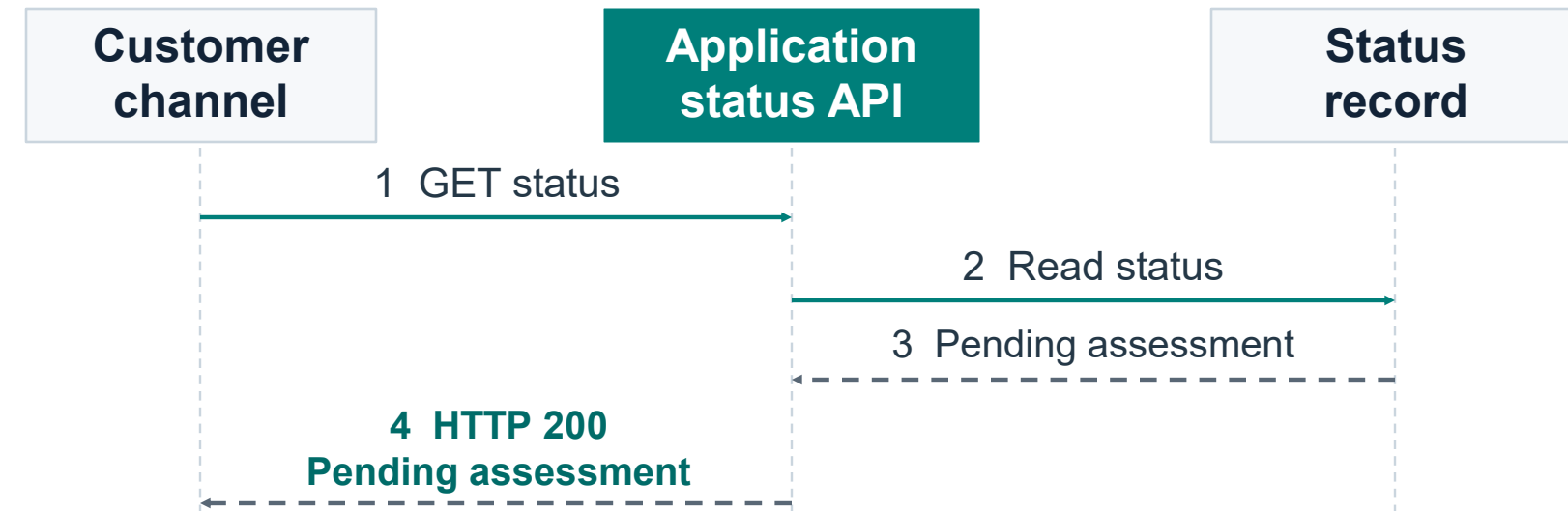
Several consumers reacting independently to the same business fact.

DESIGN WATCHPOINT

A message can be delivered again (twice, in case of failures). Handle duplicates and monitor backlog; publication does not mean every consumer has completed its work.

Request-Response

A caller sends a request and receives a corresponding response, through a synchronous exchange or an asynchronously correlated reply.



GET /applications/L-1042/status returns the recorded application status.

ILLUSTRATIVE EXAMPLE

A status query succeeds while the loan application remains Pending assessment.

GOOD FIT

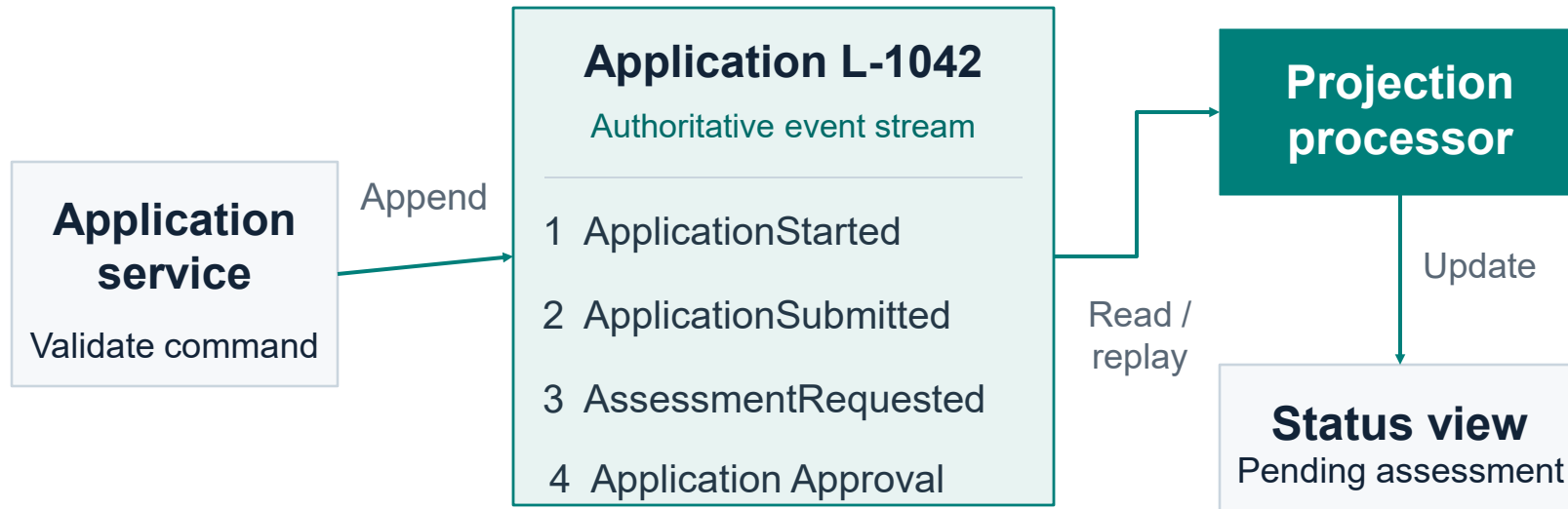
Queries or bounded operations that require a specific reply.

DESIGN WATCHPOINT

A timeout can leave an operation's outcome unknown. Define deadlines and safe retries; HTTP success does not mean loan approval.

Event Sourcing

Store state changes as an authoritative, append-only event history; reconstruct each entity's state and build read views from that ordered history.



ILLUSTRATIVE EXAMPLE

Alternative design: loan application state is derived from its own event history.

GOOD FIT

Historical reconstruction justifies the extra complexity.

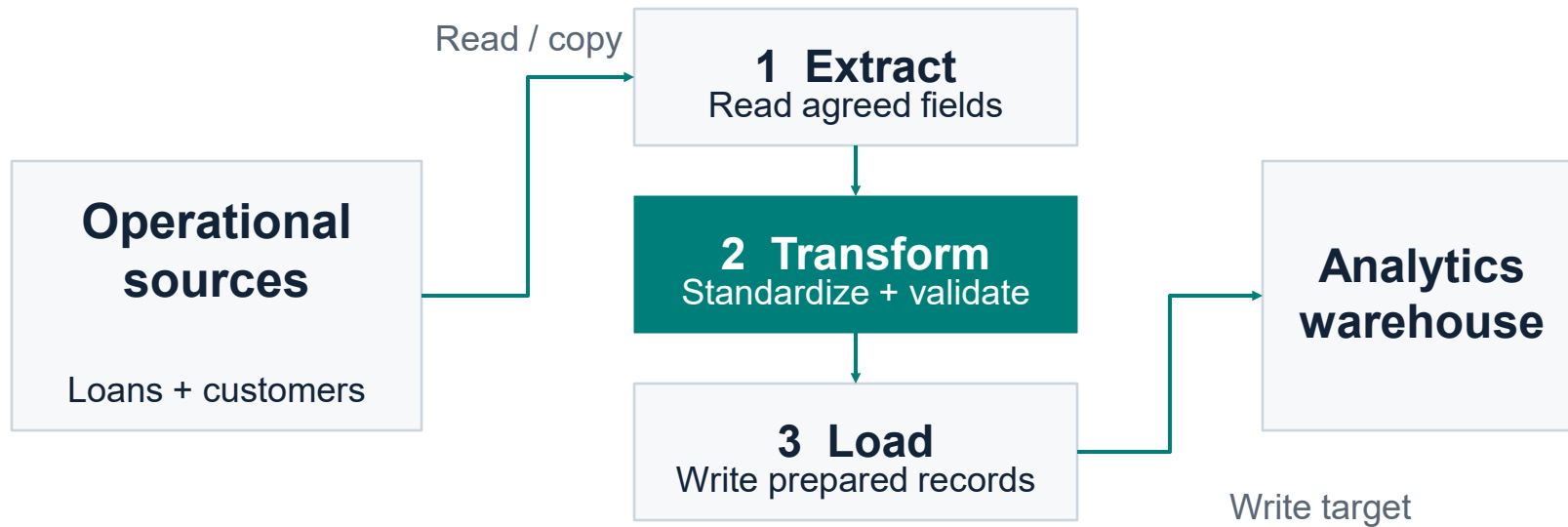
Events are authoritative for this application; read views can be rebuilt.

DESIGN WATCHPOINT

An audit log or broker alone is not event sourcing. Plan event evolution and projection lag; replay must not repeat external actions.

ETL

Extract data from source systems, Transform it into the required form, then Load it into a target repository (eg, a Data Warehouse)



Transform before loading the target. Batch timing is a separate choice.

ILLUSTRATIVE EXAMPLE

Loan and customer extracts become a consistent warehouse dataset for operational reporting.

GOOD FIT

Analytics need documented transformations and an agreed freshness target.

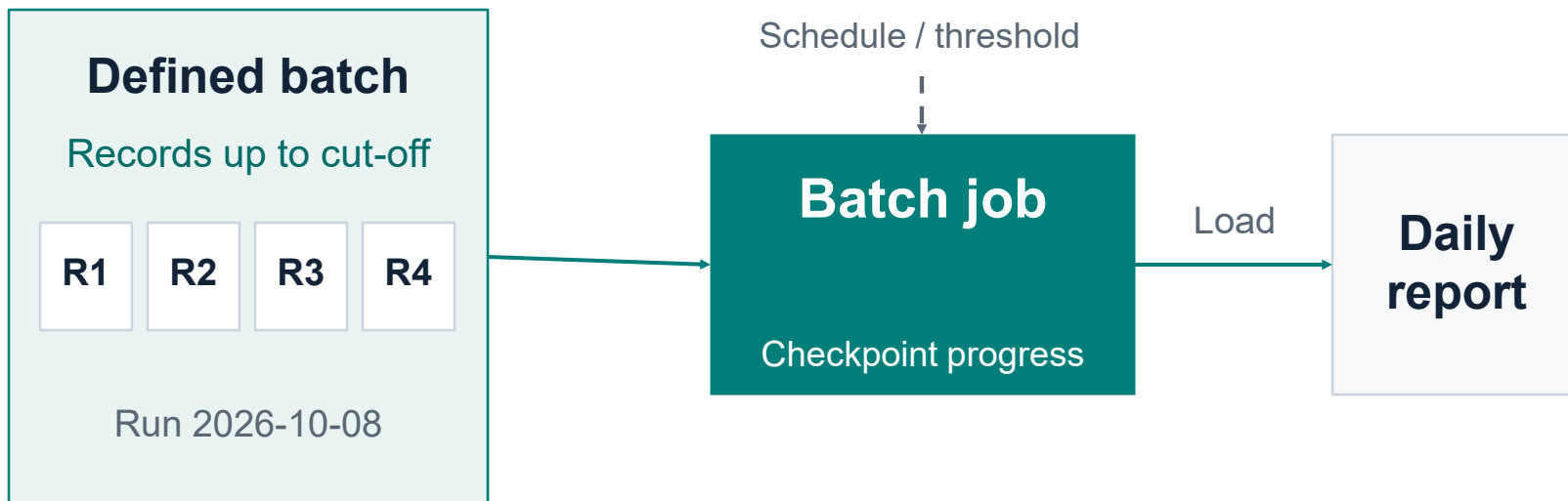
DESIGN WATCHPOINT

Solution Architecture

Define frequency, reconciliation for conflicts between different infos, lineage (of the many versions of the same data), and restart rules. ETL specifies transformation order; it does not require nightly batches or guarantee data quality.

Batching

Process a defined group of records together, starting on a schedule or when a configured collection threshold is reached.



ILLUSTRATIVE EXAMPLE

A daily reporting run processes a defined set of application records after the cut-off.

GOOD FIT

Work can wait for a schedule or a collection threshold.

A defined collection is processed as one run; internal chunks may commit separately.

DESIGN WATCHPOINT

Define restart points, late-arrival rules and duplicate-safe output. Reruns must not duplicate report rows or silently omit records.

Chain Batching

The output of the previous batch becomes the input of the next batch

Batch 1 — Import transactions

At 01:00, the bank extracts the day's loan repayment transactions from the core banking system and loads them into a staging area.

Batch 2 — Reconcile repayments

When Batch 1 finishes successfully, a second batch matches each payment against the expected instalment schedule and identifies partial or late payments.

Batch 3 — Update loan status

The output of Batch 2 triggers a third batch that updates each loan account, for example:

- Current
- 1-30 days past due
- 31-60 days past due
- Default candidate

Batch 4 — Recalculate risk indicators

Once the loan status is updated, another batch recalculates measures such as delinquency indicators, provisioning inputs or behavioural risk features.

Batch 5 — Produce downstream outputs

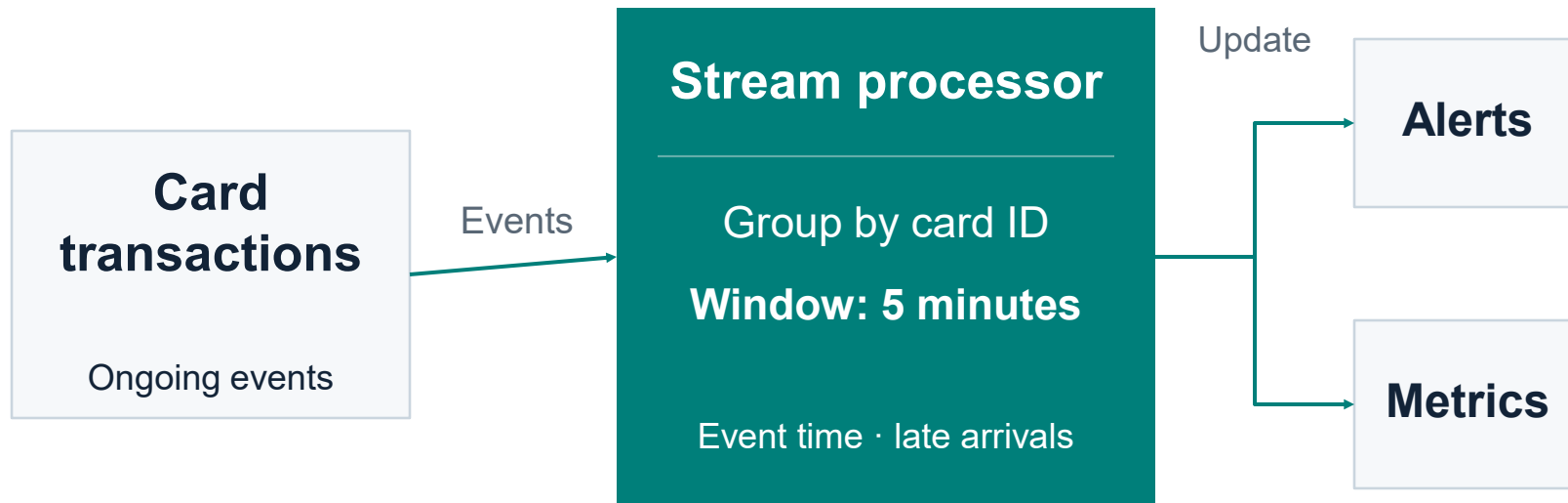
Finally, the last batch generates:

- collections worklists,
- management reports,
- regulatory extracts,
- customer-notification files.

Stream Processing

Examples: images from the bank's caveau or the bank access doors; financial info about stock exchanges; sensor / actuator updates from/to domotics, phone calls, ... Frequent stream technology is *Spark*.

Continuously process incoming events and maintain results, with explicit targets for freshness, lateness and recovery.



Windows bound an ongoing stream. Results may change when late events arrive.

ILLUSTRATIVE EXAMPLE

Card transactions are grouped by card over five minutes to flag unusual activity (fraud)

GOOD FIT

Continuous data needs timely processing as new events arrive (usually in large RAM)

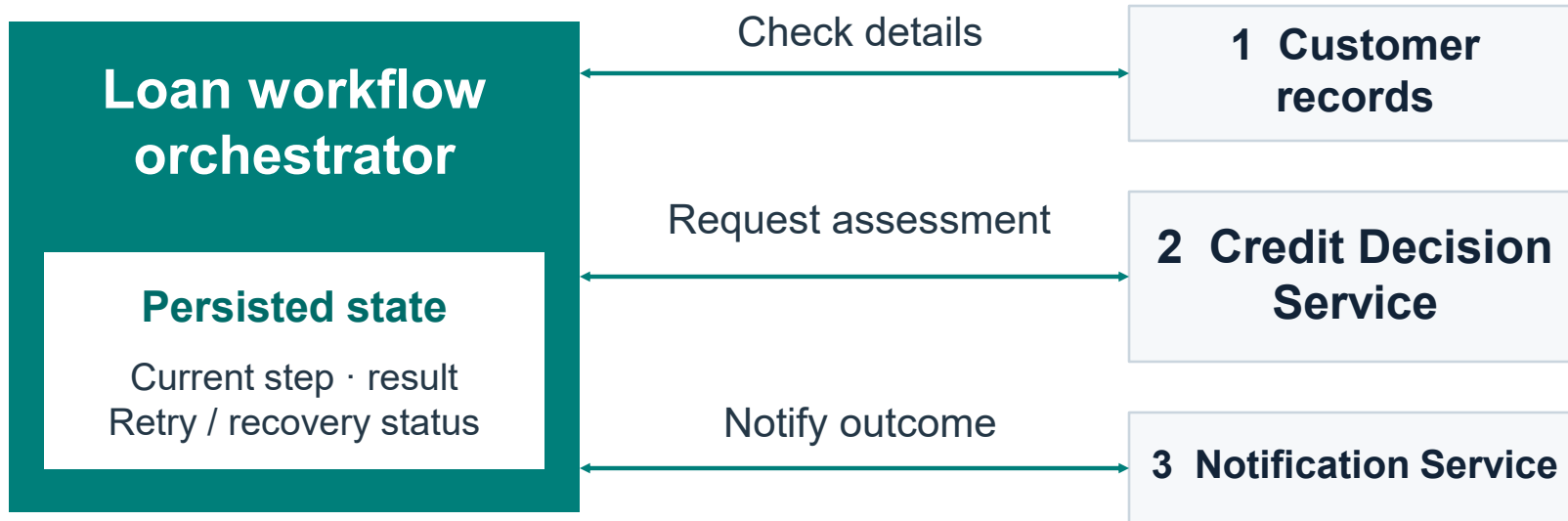
DESIGN WATCHPOINT

Set a freshness target and rules for late events. If input exceeds processing capacity, lag grows; recovery can repeat work.

The end '80 term was **workflow**
 The initial '00 term was **BPM**
 The current term is **orchestrator**.

Orchestration

Coordinate a workflow through an explicit controller that records progress, issues service commands, handles results and manages timeouts and recovery.



Workflow state selects the next step. Arrows show commands and corresponding results.

ILLUSTRATIVE EXAMPLE

Loan intake requests assessment, records progress and notifies the customer when an outcome becomes available.

GOOD FIT

A process needs an explicit owner for its sequence and recovery.

DESIGN WATCHPOINT

The coordinator owns workflow progress, not every business decision. Compensation follows business rules; it is not an automatic rollback of all completed work.